# Tyler Dong

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## EDUCATION

**Northeastern University**, B.S. Computer Science + Computer Engineering. GPA 3.976 May 2028

## EXPERIENCE

**Amazon Web Services (AWS)**, Software Development Engineer Intern (Elastic Block Store (EBS) Encryption) Jun 2026 - Aug 2026

- Re-architected a C++ disk-encryption service for static stability by removing its runtime dependency on the key service (KMS), so encrypted volumes can recover after a reboot with no network call, even when KMS is down.
- Benchmarked two NitroTPM key-sealing designs and shipped the winner: AES-256-GCM envelope encryption sealed to measured boot state. Worst-case recovery of a host's 8,000+ volume keys went from 3.3 min to about 30 ms.
- Implemented key sealing, wrapping, and root-key persistence as asynchronous state machines, validated with a Go integration suite that spins up EC2 Nitro instances to test recovery without KMS, TPM failure, and firmware tampering.

**Philips**, Software Development Engineer Co-op Jan 2026 - May 2026

- Architected a GitHub Actions orchestrator to automate migrating 15+ Azure DevOps repositories, coordinating 7 sub-workflows for repo mirroring, AI-assisted pipeline conversion, secrets injection, and variable migration.
- Built a git-tfs pipeline for legacy TFVC repositories the Azure DevOps importer couldn't handle, rewriting large binaries to Git LFS and running a daily REST API sync to keep branches current through cutover.

**Tetracorp**, Software Engineer (Bluffs) Jan 2026 - May 2026

- Owned the wallet and payments backend for Bluffs on Convex (TypeScript). Designed a three-balance ledger (deposits, withdrawable winnings, holds) where every hold remembers where its money came from, so only winnings can ever be withdrawn.
- Designed the withdrawal pipeline so users never lose money when the payout API fails: debit first, idempotent payout requests keyed to our own transaction ID, a compensating re-credit on every failure path, and webhooks matched by provider ID.

**Khoury College of Computer Sciences**, Teaching Assistant, Object-Oriented Design (CS 3100); Teaching Assistant, Discrete Structures (CS 1800) Jun 2025 - Dec 2025

- Led lab sessions and office hours 6 hours a week for 400+ students: encapsulation, inheritance, polymorphism, abstraction, MVC, and design patterns in Java.

**Outamation**, AI Automation Extern May 2025 - Jul 2025

- Built a hybrid RAG pipeline with LlamaIndex, combining dense retrieval (HuggingFace e5-large-v2) with BM25 keyword search, which doubled retrieval accuracy on complex mortgage documents. Ran Mistral-7B locally on llama.cpp.

**Quartzy Capital Advisors, LLC**, Data Research Intern Jun 2023 - Aug 2023

- Wrote a Python tool (yfinance, pandas) that automated historical financial analysis and key metrics, replacing manual Excel work and cutting processing time by 95%+.

**HomeGoods**, Merchandise Associate Aug 2022 - Aug 2025

## PROJECTS

**ETF Market Intelligence Pipeline**. Tracks every fund from 26 ETF issuers each trading day and emails a weekly brief. Mar 2026 - present

**Order Flow Imbalance Across Volatility Regimes**. Does order flow predict short-term price moves, and does that change with volatility? May 2026 - Sep 2026

**Limit Order Book Matching Engine**. A deterministic price-time priority matching engine in Java 21. Orders in, trades out. Aug - Sep 2026

**IMC Prosperity 4**. IMC's global algorithmic trading competition. Ranked 1,757th of 18,803 teams (top ~9.3%). Apr 2026

## HACKATHONS

**BostonHacks 2024**, 1st of 49, Interstellar Intelligence (AI/ML) track (SVS Lunar Client); **Civic Tech Hackathon 2025**, Best Design of 40 (AnimaGo); **HackIllinois 2025**, HackOlympian Finalist, top 5 of 105 (SpendShield); **STHacks 2025**, 4th place (ManImTired)

## SKILLS

**Languages:** Python, TypeScript, C++, Dart, Java, JavaScript, C#, Go, PowerShell, HTML/CSS

**Frameworks:** pandas, React, Next.js, FastAPI, Tailwind, Flutter, React Native, NumPy, PyTorch, Unity, LlamaIndex, Playwright

**Infra and tools:** GitHub Actions, AWS, Convex, NitroTPM, Azure DevOps, Git, Apache Arrow/Parquet, Gradle, Docker, Firebase